

Prior-Flip Diagnostics for Base-Rate Neglect in Large Language Models

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Abstract

Base-rate neglect is a classic failure mode for reasoning under uncertainty, but a prompt can appear to solve it when the evaluation confounds displayed answer options with the underlying Bayesian hypothesis. We build a train-free schema-v2 prior-flip diagnostic in which minimal pairs vary only the prior while preserving the likelihood evidence and answer format. Across 840 scored full-run rows, a hosted API model reaches a ceiling under direct, generic chain-of-thought, and prior-first prompting, while the open-weight Qwen2.5-7B-Instruct backend remains far from ceiling. The evidence therefore rejects a prior-first superiority claim: prior-first ties direct and generic chain-of-thought at 60/60 on the API run and scores 17/60 on Qwen versus 20/60 for the frequency-table condition. Positive evidence instead comes from measurement integrity and posterior availability: the schema separates option choice from Bayesian hypothesis fields with no extraction or truncation failures, and posterior-only controls outperform prior-absent controls in paired tests. The result is a diagnostic paper rather than a mitigation claim: schema-v2 scoring exposes ceiling effects, Qwen arithmetic and posterior-consistency failures, and the unsupported boundary for stronger external-baseline generalization.

Introduction

Base-rate neglect is a useful stress test for language-model reasoning because the correct answer depends on both a prior and likelihood evidence, while the tempting answer often follows representativeness alone (Tversky and Kahneman 1974; Bar-Hillel 1980). Classic work also shows

that frequency representations can change Bayesian reasoning outcomes without changing the underlying information (Gigerenzer and Hoffrage 1995). For LLMs, this means that a final multiple-choice answer is not enough: a model may choose the displayed option, mention a base rate, or write a posterior-like number without consistently tracking the Bayesian hypothesis.

Recent LLM work reinforces that measurement problem. Probabilistic-reasoning studies show that fluent rationales do not guarantee coherent belief or posterior behavior (Li and Rui 2024; Hobor et al. 2026; Freedman and Toni 2025). Cognitive-bias benchmarks show that prompt format, answer format, and benchmark construction can change measured bias rates (Echterhoff et al. 2024; Shaikh et al. 2024; Malberg et al. 2024). Prompt-scaffold and Bayesian-teaching studies motivate inference-time structure, but they also make comparator choice important: a train-free scaffold should be credited only when it improves over serious train-free alternatives (Araya 2025; Qiu et al. 2025; Sun et al. 2025).

The gap addressed here is the controlled prior-flip case. Prior work motivates asking whether models use priors and whether posterior-like answers are coherent, but it less directly isolates minimal pairs in which the likelihood evidence and displayed answer format stay fixed while only the prior changes. That isolation turns base-rate evaluation into a paired measurement problem: a prior-sensitive model should change its answer when the base rate changes, and the evaluation should record whether the displayed option, latent

Bayesian hypothesis, and posterior explanation agree.

We therefore build a train-free schema-v2 prior-flip diagnostic with 30 semantic pairs, seven prompt conditions, two evaluated backends, and exact paired tests. The prompt conditions include direct prompting, generic chain-of-thought, frequency-table prompting, prior-first prompting, posterior-only, prior-blind, and likelihood-only controls. The scorer separates displayed option choice, inferred Bayesian hypothesis, posterior numeric consistency, arithmetic failure, extraction failure, and truncation failure before aggregation. Figure 1 summarizes this response-to-fields-to-claims path.

Across 840 scored rows, the evidence supports a measurement contribution rather than a prior-first mitigation claim. The hosted API backend is saturated: direct, generic chain-of-thought, prior-first, and posterior-only all reach 60/60. The open-weight Qwen backend is non-ceiling, but prior-first scores 17/60, below the 20/60 frequency-table baseline. Posterior-only controls improve over prior-absent controls, and the Qwen failure profile is dominated by arithmetic and posterior-consistency errors rather than extraction or truncation. The contributions are a prior-flip benchmark, a schema-v2 field protocol for auditing answer/hypothesis/posterior agreement, and a calibrated empirical result showing where the planned prior-first claim fails and which diagnostic signals remain supported.

Related Work

Classic work on heuristics and biases established that base rates are often neglected when representativeness is salient (Tversky and Kahneman 1974; Bar-Hillel 1980). Frequency-format interventions are especially relevant to this paper because they change the representation of Bayesian information without training a model (Gigerenzer and Hoffrage 1995). We therefore treat frequency-table prompting as a strong baseline for a train-free prior-first scaffold.

LLM probabilistic-reasoning work studies whether model outputs obey coherent belief, posterior, or uncertainty con-

straints (Li and Rui 2024; Hobor et al. 2026; Freedman and Toni 2025). This paper follows that line by measuring posterior consistency and arithmetic failures, but it narrows the task to base-rate prior flips so that paired comparisons are available. Cognitive-bias benchmark work motivates broader task families and public validation (Echterhoff et al. 2024; Zhu, Yan, and Griffiths 2025; Shaikh et al. 2024; Malberg et al. 2024). Our current evidence covers one controlled benchmark family, so the paper reports benchmark-family scope as a limitation instead of turning those citations into unrun external results.

Prompting and Bayesian-teaching studies motivate structured inference-time support (Araya 2025; Qiu et al. 2025; Sun et al. 2025). The present evaluation differs in two ways. It is train-free and compares against direct, generic reasoning, and frequency-table baselines; and it treats posterior-only as a diagnostic control rather than as a fair user-facing scaffold. This positioning lets the paper ask whether the motivating prior-first intervention survives a strong minimal-pair audit. In the current evidence, it does not.

Method and Benchmark Design

The benchmark is a base-rate prior-flip dataset with 30 semantic pair identifiers. Each item has paired prior variants, balanced displayed option sides, and balanced correct Bayesian hypothesis sides. Seven prompt conditions are evaluated for each backend, giving 420 scored rows per backend and 840 scored rows overall. The design is inference-only: no model weights, prompts, or scorers are trained.

The harness turns each semantic pair into matched prior variants and prompt-condition rows. Only the prior changes inside a minimal pair; the likelihood evidence and displayed answer format remain fixed. This makes base-rate use a paired measurement rather than a collection of independent examples. A prompt condition receives credit only when its extracted displayed option matches the option implied by the Bayesian hypothesis for that row.

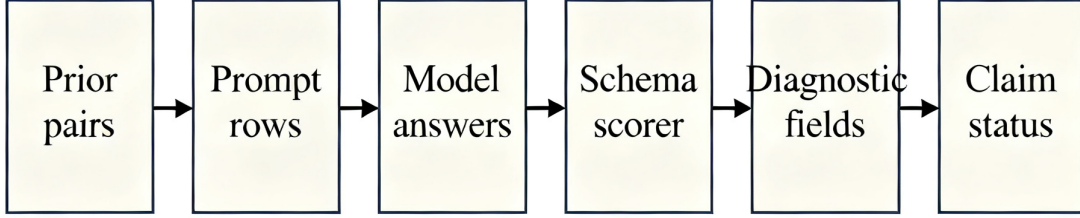


Figure 1: Schema-v2 prior-flip diagnostic: paired priors become prompt rows, responses are parsed into fields, and only table-backed aggregates determine claims.

Operationally, one evaluation row has four reader-facing steps. First, the benchmark row specifies the semantic item, prior variant, likelihood evidence, displayed options, expected Bayesian hypothesis, and prompt condition. Second, the selected backend receives the corresponding prompt under the stated deterministic decoding budget and returns a text response. Third, the schema-v2 scorer extracts the displayed option, inferred Bayesian hypothesis, any posterior value, arithmetic and posterior-consistency flags, and parser-quality flags. Fourth, aggregate tables and exact McNemar tests are computed from those fields before any claim is written.

The scorer is a rule-based schema-v2 parser. It extracts displayed option choice, inferred Bayesian hypothesis, posterior numeric consistency, arithmetic failure, extraction failure, truncation failure, format noncompliance, prior mention, and evidence-overweight error. These fields let the paper distinguish a correct final option from a coherent posterior explanation, and they keep parser quality separate from reasoning failures.

The prompt conditions are direct, generic chain-of-thought, frequency-table, prior-first, posterior-only, prior-blind, and likelihood-only. Direct, generic reasoning, and frequency-table prompting are baselines; prior-first is the candidate scaffold; posterior-only and prior-absent variants are diagnostic controls. Frequency-table is treated as a serious train-free baseline because the literature on natural frequencies makes it more than a cosmetic prompt variant.

The body keeps one conceptual overview and three data figures so that the quantitative evidence is visible where the corresponding claims are discussed. The appendix records protocol details and compact evidence ledgers rather than repeating the body visuals.

Experimental Setup

The measured benchmark family is schema-v2 prior flip: 30 semantic pairs, two prior variants per pair, seven train-free prompt conditions, and 420 scored rows per backend. The evaluated backends are a hosted ‘gpt-5-mini’ API run and Qwen2.5-7B-Instruct, both using deterministic decoding, the same prompt definitions, and the same rule-based scorer. The prompt conditions are direct, generic reasoning, frequency-table, prior-first, posterior-only, prior-blind, and likelihood-only; the main metric is base-rate choice accuracy, with exact McNemar tests on matched rows. The API run used a 160-token output budget and the Qwen run used a 220-token output budget.

One benchmark episode is deliberately non-agentic. The row controller instantiates one schema-v2 prior-flip item, selects one prompt condition, sends the prompt to the evaluated backend, and hands the raw text to the parser without search, tool use, memory, self-repair, or learned adaptation. The parser extracts the displayed option, the mapped Bayesian hypothesis, any stated posterior, arithmetic and posterior-consistency flags, and response-quality gates. Those fields

are the only inputs to the scorer; condition aggregates and McNemar tests are then computed from matched scored rows. Appendix Table 3 records the same protocol in scan-friendly form after the references, while the body proceeds directly from setup prose to the main result matrix.

Results

Table 1 gives the full cross-condition matrix and is the primary body result anchor. The API backend is ceiling-saturated: direct, generic chain-of-thought, prior-first, and posterior-only all reach 60/60, while frequency-table is nearly saturated at 59/60. This makes an API-side prior-first improvement unmeasurable in the current split. Qwen is not saturated: total base-rate choice accuracy is 138/420, and prior-first scores 17/60, below the 20/60 frequency-table condition.

The matrix should be read as a claim filter. A high API prior-first score supports the statement that this backend can solve the measured rows under that prompt, but it does not support a method advantage because direct and generic reasoning are also perfect. The Qwen rows provide a harder regime, yet there the candidate prior-first scaffold remains below the frequency-table baseline.

The ceiling regime is important because it blocks a common but invalid interpretation. If prior-first were inspected alone on the hosted backend, 60/60 would look like a successful intervention. The matched comparator rows change that conclusion: direct prompting and generic chain-of-thought also reach 60/60, while frequency-table is 59/60. The appropriate claim is therefore not that prior-first improves the hosted backend, but that this split is too saturated to measure a hosted-backend prompt advantage.

The Qwen regime provides the non-ceiling check. Qwen remains below perfect accuracy in every condition, so a useful prompt scaffold has room to separate itself. It does not: prior-first reaches 17/60, while frequency-table reaches 20/60 and posterior-only reaches 33/60. This pattern keeps the

negative result from being an artifact of saturation alone. The candidate scaffold fails in the regime where improvement would have been observable.

The posterior-only row should not be read as a rescued method. It is a diagnostic control that changes information availability, not a fair user-facing prior-first scaffold. Its advantage over prior-blind and likelihood-only rows shows that the benchmark can register a useful information channel when one is supplied. The same matrix therefore supports a positive control result and rejects the planned mitigation claim.

Two consequences follow from making the matrix the body anchor. First, every headline sentence has a denominator: the relevant unit is 60 rows per condition per backend, not an unqualified percentage. Second, comparator choice is visible. Prior-first is compared with direct, generic reasoning, and frequency-table baselines, while posterior-only is compared with prior-absent controls. These comparator roles are what prevent a ceiling score or an ablation win from being over-read as a deployed scaffold result.

Analysis, Ablation, and Failure Cases

The paired tests in Table 2 reject prior-first superiority. API prior-first ties direct and generic chain-of-thought exactly, with McNemar $p = 1.0$. Qwen prior-first also fails to beat frequency-table, with $p = 0.375$. Posterior-only beats prior-absent controls for both backends, so the supported positive result is an information-availability diagnostic, not a deployable prior-first method.

Posterior-only is deliberately not framed as a fair user-facing scaffold. It supplies information that the original task asks the model to infer, so its role is diagnostic. Its paired gains show that the benchmark can register an information-availability effect; they do not rescue the prior-first method claim. This distinction matters because otherwise the strongest positive ablation could be misread as the proposed intervention.

Table 1: Main result: API is ceiling-saturated while Qwen prior-first does not beat frequency-table.

Condition	Role	API acc.	API prior sens.	Qwen acc.	Qwen post. cons.	Qwen arith. fail
Direct	Un scaffolded baseline	100.0%	100.0%	28.3%	23.3%	76.7%
Generic CoT	Reasoning baseline	100.0%	100.0%	28.3%	35.0%	65.0%
Frequency table	Frequency baseline	98.3%	96.7%	33.3%	21.7%	78.3%
Prior first	Candidate scaffold	100.0%	100.0%	28.3%	20.0%	80.0%
Posterior only	Posterior control	100.0%	100.0%	55.0%	0.0%	100.0%
Prior blind	Prior-absent control	43.3%	10.0%	31.7%	1.7%	98.3%
Likelihood only	Likelihood-only control	43.3%	10.0%	25.0%	1.7%	98.3%

Table 2: Paired tests: prior-first superiority is unsupported, but posterior-only controls beat prior-absent controls.

Run	Comparison	pairs	A correct/B wrong	A wrong/B correct	exact p
API	prior first vs direct	60	0	0	1
API	prior first vs generic cot	60	0	0	1
API	prior first vs frequency table	60	1	0	1
API	posterior only vs prior blind	60	34	0	1.16e-10
API	posterior only vs likelihood only	60	34	0	1.16e-10
Qwen	prior first vs direct	60	1	1	1
Qwen	prior first vs generic cot	60	1	1	1
Qwen	prior first vs frequency table	60	1	4	0.375
Qwen	posterior only vs prior blind	60	19	5	0.00661
Qwen	posterior only vs likelihood only	60	19	1	4.01e-05

Figure 3 visualizes the same matched discordance counts from Table 2. The left/right bars make the negative prior-first result easier to audit: the selected prior-first comparisons do not produce a systematic matched-row advantage, while posterior-only controls produce discordant wins against prior-absent controls.

Qwen’s failure profile explains why the open-weight backend is diagnostically useful. It has 358 arithmetic-failure rows out of 420 and only 62 posterior-numeric-consistent rows, with no extraction or truncation failures in the active full rows. Figure 4 decomposes those failures by condition.

The failure taxonomy narrows the next intervention target. A low Qwen score could have been caused by extraction loss, truncation, or answer-format noncompliance; those gates are zero in the full rows. The remaining pattern is arithmetic

and posterior-consistency failure, so a future scaffold should target computation and posterior expression rather than only telling the model to inspect the prior first.

Discussion and Scope

The evidence supports a diagnostic story rather than a mitigation story. In the ceiling regime, the hosted backend answers every measured row correctly under direct, generic reasoning, and prior-first prompts, so a prior-first effect is unmeasurable. In the non-ceiling regime, Qwen exposes prompt differences, but prior-first still does not outperform frequency-table.

The strongest limitation is source-family scope. The full evidence uses one controlled benchmark family, one hosted backend, and one open-weight backend. That is enough to test the minimal-pair question and reject the active prior-first

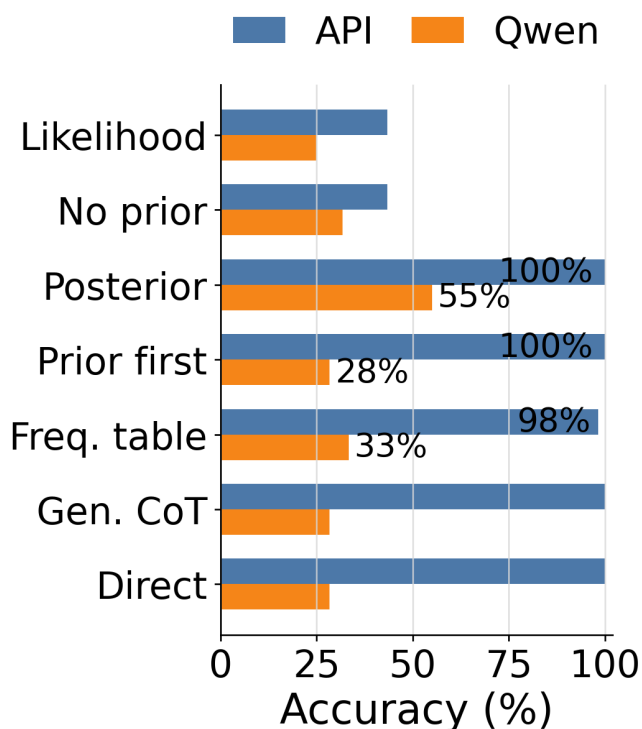


Figure 2: Per-condition accuracy from the full schema-v2 runs shows API ceiling behavior and non-ceiling Qwen failures.

superiority claim, but not enough for broad cognitive-bias or model-family generalization (Tversky and Kahneman 1974; Bar-Hillel 1980; Gigerenzer and Hoffrage 1995; Li and Rui 2024; Hobor et al. 2026; Echterhoff et al. 2024; Zhu, Yan, and Griffiths 2025; Shaikh et al. 2024; Qiu et al. 2025). External baselines, harder splits, and model sweeps remain future validation cells until comparable raw outputs, scored rows, aggregate tables, and paired tests exist.

Evidence Trace and Threats

The evidence trace has three levels: row-level schema-v2 fields, condition-level aggregates, and matched claim comparisons. This separation is why an API prior-first score of 60/60 is not reported as a method win: direct and generic chain-of-thought also score 60/60. It is also why posterior-only remains an information-availability control rather than a proposed scaffold.

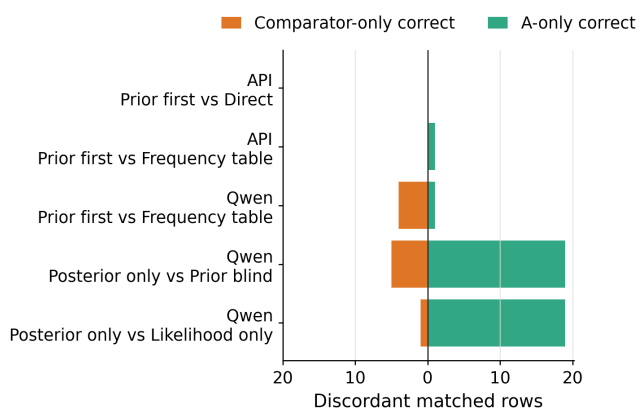


Figure 3: Paired discordance summary: selected matched comparisons show no prior-first advantage over direct or frequency-table baselines, while Qwen posterior-only controls gain discordant wins against prior-absent controls.

The main threats are ceiling saturation, narrow source-family scope, model scope, and intervention scope. Qwen2.5-7B-Instruct is one backend, not a model-family sample, and the current matrix lacks completed CANNOT, BiasBuster, Bayesian Teaching, harder-split, and broader model-sweep rows. Those cells are future validation targets, not hidden evidence.

The row fields also explain why the paper treats measurement integrity as a result. The scored rows have no extraction failures and no truncation failures. That means the negative prior-first result is not caused by missing answer extraction or cut-off responses. The remaining errors are model-output errors under the tested prompts, which is the behavior the diagnostic is meant to expose.

The posterior field gives a second check. A model can choose the displayed option while writing a posterior that does not match the task values. It can also mention the prior without using it in the final hypothesis. Schema-v2 keeps those cases visible, so the analysis reports posterior numeric consistency and arithmetic failure next to accuracy rather than hiding them in a separate appendix.

The paired tests provide the third check. They ask whether one condition wins on the same semantic items where an-

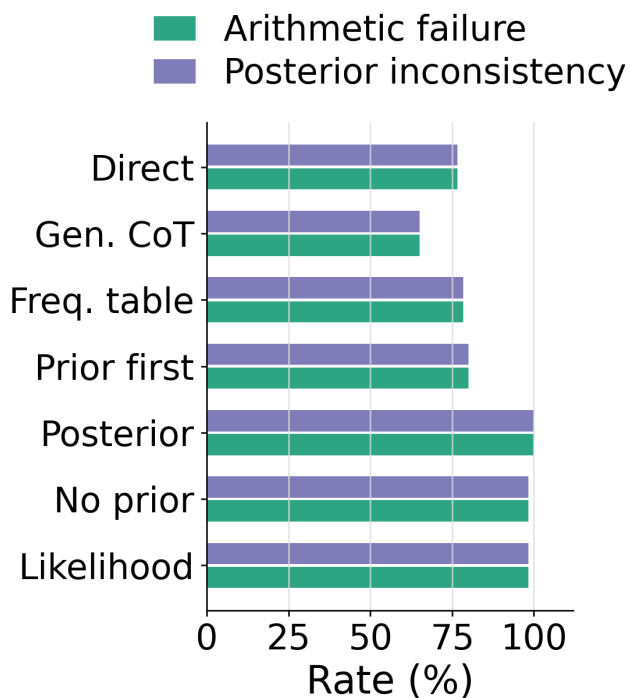


Figure 4: Qwen failure decomposition by prompt condition shows arithmetic and posterior-consistency failures rather than extraction or truncation failures.

other condition loses. This matters for the Qwen prior-first comparison because aggregate accuracy alone could make a small difference look meaningful without matched-row support. The exact counts show that prior-first does not have the systematic advantage needed for the planned scaffold claim.

The frequency-table condition is kept in the main comparison because it is a serious train-free representation baseline, not a decorative prompt variant. In the non-ceiling Qwen run it outperforms prior-first, which is why the candidate scaffold is not evaluated only against direct prompting. A mitigation claim should survive direct, generic reasoning, and frequency-format comparators on matched rows.

The API ceiling result is useful for the opposite reason. A saturated backend can produce a perfect prior-first score without evidence of a method advantage. The correct conclusion is not that all prompts are equivalent in general, but that this split cannot identify a small prompt gain for that

backend. Future harder splits should preserve the prior-flip invariant while choosing priors and likelihoods that avoid ceiling behavior.

The Qwen result identifies a concrete repair target. The active full rows are parser-clean, while arithmetic failure and posterior inconsistency dominate the failure taxonomy. A future scaffold should therefore target computation and posterior expression, not only the instruction to inspect the prior first.

The body figures make these checks inspectable without adding new claims. Figure 2 separates the saturated hosted regime from the non-ceiling Qwen regime. Figure 3 shows the matched comparisons that reject prior-first superiority. Figure 4 then localizes the non-ceiling failures to arithmetic and posterior-consistency fields.

Posterior-only controls are useful because they show what the task can register when posterior information is supplied. They are also easy to overread. Since posterior-only changes the information available to the model, it cannot be treated as evidence that the model inferred the posterior from the original prior and likelihood. The paper keeps that result as a diagnostic control and not as the proposed method.

The distinction between missing evidence and contrary evidence is central to the review claim. External benchmark cells are missing and should be added before broader conclusions are written. Prior-first superiority is different: the active rows speak against that planned claim because the candidate scaffold ties saturated API baselines and trails frequency-table prompting on Qwen.

This distinction keeps the measured negative from collapsing into a null paper. The protocol exposes a failed planned scaffold, but it also preserves useful positive and diagnostic signals: posterior availability matters, parser-quality gates are clean, and Qwen errors have an observable arithmetic/posterior profile. Those signals define the next experiment more precisely than a single aggregate score would.

The allowed body claim is narrow: schema-v2

prior flips expose ceiling effects, a non-winning prior-first scaffold, posterior-availability controls, and Qwen arithmetic/posterior-consistency failures. The appendix gives the longer replay protocol and claim-boundary ledger; the body keeps the tables and figures as the claim authority.

Conclusion

Under the current full schema-v2 evidence, prior-first prompting should not be presented as a winning mitigation for base-rate neglect. The supported contribution is a diagnostic one: separating option choice from Bayesian hypothesis fields exposes API ceiling effects, Qwen arithmetic and posterior-consistency failures, and the value of posterior-available controls. The next evidential step is not prose polish but new audited cells for harder splits and external baselines.

Limitations

The study is intentionally narrow: one base-rate prior-flip family, two evaluated backends, and seven train-free prompt conditions. It rejects the active prior-first superiority claim but does not estimate base-rate neglect across cognitive-bias families, model families, or deployment settings. Posterior-only is an information-availability ablation, not a deployable mitigation; CANNOT, BiasBuster, Bayesian Teaching, harder-split, and model-sweep cells remain future validation targets until comparable raw outputs, scored rows, aggregate tables, and paired comparisons exist.

The protocol is also output-level. It records displayed options, mapped hypotheses, posterior expressions, arithmetic flags, and parser-quality gates, but it does not inspect internal model states or identify causal mechanisms. The Qwen failure taxonomy should therefore be read as a description of observed responses under this prompt set, not as a claim about hidden computation or all open-weight models.

A stronger paper would add evidence before broadening the story. The most direct extensions are a harder prior-flip split that reduces hosted-backend ceiling saturation, external cognitive-bias cells with compatible schema fields, and a

model sweep that keeps prompt definitions and parser policy fixed. Each extension would need refreshed aggregates, paired tests, figures, captions, and claim boundaries.

The current split also leaves task difficulty underexplored. It shows that the hosted backend can saturate a controlled prior-flip family, but it does not identify the prior and likelihood ranges where that backend begins to fail. A harder split should vary those values prospectively while preserving the minimal-pair invariant, balanced displayed-option sides, and balanced Bayesian-hypothesis sides.

External validation has a separate comparability limit. Some cognitive-bias datasets provide only final multiple-choice labels, while this protocol also depends on mapped hypotheses, posterior expressions, arithmetic fields, and parser-quality gates. Such datasets can still be useful, but they would support a narrower final-choice claim unless additional elicitation or schema mapping is added.

Model scope is similarly limited. The two evaluated backends demonstrate two regimes in the active matrix, not a distribution over model families. A model sweep would need comparable decoding settings, parser-quality checks, and per-backend paired comparisons before the paper could replace the present diagnostic language with a broader model-family statement.

Responsible Use

The benchmark is an audit instrument, not evidence that any deployed model is safe for high-stakes probabilistic decisions. The main foreseeable misuse is rhetorical: citing the perfect API prior-first row as a mitigation result, or citing the Qwen failure row as a broad model-family defect. Both readings are outside the measured evidence. Application-facing use would need domain tasks, domain priors and likelihoods, calibrated harm definitions, and independent validation under the same paired-comparison discipline.

For high-stakes settings, the diagnostic should be treated as a pre-deployment research probe only. A domain eval-

uation would need domain-specific priors and likelihoods, prospective task construction, independent scoring checks, and a decision policy that accounts for the cost of wrong posterior reasoning. The present result can motivate such audits, but it does not certify any system for consequential decisions.

The same caution applies to human-facing explanations. A fluent model rationale can make an answer look Bayesian even when the displayed option, mapped hypothesis, posterior expression, and arithmetic field disagree. Any downstream use should therefore preserve the field-level audit rather than replacing it with a single natural-language confidence statement.

The diagnostic is also not a recommendation to hide uncertainty from users. If a system is evaluated with this protocol, the report should state the benchmark family, the prompt conditions, the evaluated backend, the denominator, and the paired-test comparator. Reporting only a final accuracy number would erase the distinction between ceiling saturation, a failed scaffold comparison, and an information-availability control.

Finally, the paired design should not be used as an after-the-fact filter for favorable cases. The semantic pairs, prior variants, prompt conditions, parser fields, and paired-test policy need to be fixed before model responses are interpreted. Otherwise a positive result could reflect item selection rather than a real change in base-rate-sensitive behavior.

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Reproducibility Checklist

The evaluated protocol records the source family, two backends, seven prompt conditions, schema-v2 parser fields, condition-level result table, paired-significance table, failure taxonomy, and generated data figures. Paper-facing claims use only completed full-run rows and paired tests.

Appendix: Supplementary Evidence

This supplement records the evidence boundary without adding new result claims. The active matrix has one benchmark family, one hosted backend, one open-weight backend, seven prompt conditions, and 840 scored rows. External CANNOT, BiasBuster, Bayesian Teaching, harder-split, and model-sweep cells are absent from the result matrix and are treated as future validation targets.

Protocol Summary

The supplementary protocol summary keeps metadata and outcomes in one compact evidence table. Condition labels are defined in the next subsection so the table can stay short.

Prompt Conditions and Metrics

Direct prompting is the unscaffolded baseline. Generic chain-of-thought asks for reasoning without forcing prior order. Frequency-table prompting is a strong train-free representation baseline. Prior-first is the candidate scaffold. Posterior-only, prior-blind, and likelihood-only are diagnostic controls. The scorer records displayed option, Bayesian hypothesis, posterior numeric consistency, arithmetic failure, extraction failure, truncation failure, format noncompliance, and evidence-overweight error.

Base-rate choice accuracy answers whether the displayed option matches the answer implied by the Bayesian hypothesis for that row. Posterior numeric consistency asks whether a stated posterior agrees with the prior and likelihood values in the item. Arithmetic failure marks responses where the calculation itself is incoherent or incompatible with the stated evidence. Extraction failure, truncation failure, and format noncompliance are response-quality gates rather than cognitive-bias findings.

The compact comparator table keeps only the row status. The decision rule is stated in prose: a prior-first prompt can support a gain claim only if matched rows separate it from direct prompting, generic reasoning, and the frequency-table representation (Gigerenzer and Hoffrage 1995). The

current matrix does not show that pattern, so the abstract and conclusion report a diagnostic negative result rather than a mitigation claim. Posterior-only is interpreted separately because it changes information availability instead of testing the original user-facing task.

Claim Boundary and Reader Check

Supported claims are the full evidence size, API ceiling behavior, Qwen diagnostic failures, posterior-only control gains, schema-v2 measurement integrity, and Qwen arithmetic/posterior-consistency failure decomposition. The rejected claim is prior-first superiority over strong train-free baselines. Missing or weak claims include external-baseline generalization, harder-split validation, and broad model-family conclusions.

A reader can check the manuscript in five steps. First, compare the abstract’s negative prior-first sentence with Tables 1 and 2. Second, check that every prompt condition named in the prose appears in the main matrix. Third, verify that posterior-only is described as a diagnostic control rather than as the proposed method. Fourth, inspect Figure 4 to see whether Qwen failures are parser failures or reasoning-output failures. Fifth, confirm that external benchmarks are named only as missing validation cells.

Evidence Walkthrough

The result trace begins with paired prior-flip rows. Each semantic pair has matched prior variants, balanced displayed answer sides, and balanced Bayesian hypothesis sides. The row then enters one of seven prompt conditions for one evaluated backend. The scoring layer extracts the displayed answer option, maps it to the task hypothesis labels, records posterior numeric consistency when a posterior is stated, and marks arithmetic, extraction, truncation, format, prior-mention, and evidence-overweight fields.

At the condition level, the aggregate table reports a denominator of 60 rows for each backend-condition cell. API ceiling behavior is visible because direct, generic reason-

Table 3: Setup and outcome matrix: both backends use the same seven-condition schema-v2 protocol, but API saturates while Qwen prior-first remains below frequency-table.

Source	Backend	Rows	Conditions	Metric	Headline
Schema-v2 pairs	prior-flip API ('gpt-5-mini')	420	seven prompts; shared scorer	Acc.; McNemar	direct/prior-first 60/60
Schema-v2 pairs	prior-flip Qwen2.5-7B-Instruct	420	same prompts and scorer	Acc.; diagnostics	prior-first 17/60; frequency 20/60
Qwen control slice	same backend	60	posterior-only	Acc.; diagnostics	posterior-only 33/60

Table 4: Comparator and claim-status summary: prompt roles, decision cues, and admitted status are separated from the surrounding interpretation prose.

Evidence item	Protocol role	Decision cue	Active status
Direct / Generic CoT	reference prompts	matched baseline	API ceiling
Frequency table	representation baseline	strongest Qwen comparator	prior-first trails
Prior first	candidate scaffold	paired method test	superiority rejected
Posterior only	availability control	channel sensitivity	positive control
Prior-blind / likelihood-only	information ablations	incomplete evidence	lower controls
API ceiling regime	backend regime	saturated rows	no gain claim
Qwen non-ceiling regime	failure surface	parser-clean errors	diagnostic claim
External benchmark cells	missing validation	no scored rows	future work

ing, prior-first, and posterior-only all have 60 correct rows. Frequency-table has 59 correct rows. Qwen behavior is different: no condition reaches ceiling, posterior-only has the highest base-rate choice accuracy, and prior-first remains below frequency-table. The paper therefore avoids averaging the two backends into a single number that would hide the central contrast.

At the comparison level, exact McNemar tests use matched rows rather than independent proportions. On the API backend, prior-first has no discordant wins over direct or generic reasoning. On Qwen, the prior-first versus frequency-table comparison has fewer prior-first-only wins than frequency-table-only wins. These paired counts are the reason the conclusion rejects prior-first superiority even though prior-first

has a high absolute API score.

Posterior-only has a different interpretation. It is a positive diagnostic control because it shows what happens when posterior information is supplied directly. Its paired wins over prior-blind and likelihood-only conditions demonstrate an information-availability effect. The paper does not call this a user-facing mitigation, because the task originally asks the model to infer the posterior from the prior and likelihood evidence.

The matched-comparison rule is now folded into Table 4. On the API backend, prior-first has no discordant advantage over direct or generic reasoning because the split is already saturated. On Qwen, the prior-first versus frequency-table comparison is the relevant fair scaffold test, and it does not

favor prior-first. Posterior-only comparisons are interpreted separately as information-availability controls.

The Qwen failure taxonomy explains what the non-ceiling regime exposes. The active rows have no extraction or truncation failures, so low accuracy is not attributed to missing parser output or cut-off responses. Arithmetic failure and posterior inconsistency dominate instead. A future scaffold that only changes the order of prose reasoning may therefore be insufficient; a stronger intervention should target computation and posterior expression while still competing against frequency-table prompting.

This interpretation uses the schema fields summarized later in Table 7. The important distinction is between parser-quality fields and reasoning-output fields. Extraction, truncation, and format-compliance flags ask whether a usable response exists. Displayed option, hypothesis mapping, posterior expression, and arithmetic consistency ask what the usable response says. The active Qwen rows are parser-clean, so the failure discussion focuses on posterior and arithmetic fields rather than missing responses.

External-Cell Readiness

An external benchmark cell becomes paper-ready only after the benchmark source is accessible, the use terms are recorded, the task fields can be mapped to displayed option and Bayesian hypothesis labels, raw model outputs exist, and scoring produces row-level fields comparable with schema-v2. If any of those steps is missing, the cell can appear in future-work planning but not in the result matrix.

The same standard applies to a harder split inside the current benchmark family. The split should be designed before seeing model outputs, should preserve balanced option sides and balanced hypothesis sides, and should keep the prior as the manipulated variable. Difficulty should not be obtained by parser-hostile wording or by selecting failures after the fact. A harder split that preserves these constraints would directly address the API ceiling limitation.

For a new prior-first claim, the report should say which

comparator is being beaten and on how many matched rows. For a new posterior-control claim, it should say which information was added and which prior-absent condition is the comparator. For a new external benchmark claim, it should say which schema-v2 fields are preserved and which are missing. A prompt name alone is not a method result; a high accuracy number without a comparator is not a superiority result; a famous benchmark name without completed scored rows is not external validation.

Supplementary Row Construction Notes

The row construction notes are included to make the minimal-pair claim inspectable without adding new experiments. Each semantic item begins with a situation in which likelihood evidence is salient enough to tempt a representativeness-style response. That design choice follows the base-rate-neglect literature: the diagnostic question is not whether a model can repeat a prior, but whether it combines the prior with evidence when the evidence description is attention-grabbing (Tversky and Kahneman 1974; Bar-Hillel 1980). The row pair then changes the base rate while preserving the semantic story and likelihood wording.

The prior variant is the manipulated variable. If the likelihood evidence also changed, the pair would no longer isolate prior use. If the answer options changed their wording at the same time, the pair could measure lexical preference. If the correct Bayesian hypothesis always appeared on the same side, the pair could measure option-side bias. The schema-v2 fields exist to keep those confounds visible. The displayed option field records the surface answer. The Bayesian-hypothesis field records what that answer means for the row. The paired analysis then asks whether the mapped hypothesis moves with the prior.

The prompt-condition expansion is mechanical once a row pair exists. Direct prompting, generic reasoning, frequency-table prompting, prior-first prompting, posterior-only, prior-blind, and likelihood-only all use the same underlying row identity. This prevents the candidate condition from receiving

a different item distribution than its comparators. It also makes the negative result more informative: prior-first is not losing because it was evaluated on a different task set from frequency-table prompting.

The posterior-only control deliberately violates the original information constraint, which is why the paper does not call it a mitigation. Its role is to check whether the evaluation can register a positive signal when the relevant posterior channel is made available. This matters because a pure negative result can be ambiguous: the candidate scaffold may fail, or the benchmark may be insensitive. Posterior-only helps separate those cases. It shows that the benchmark can move when a strong information channel is supplied, while still leaving prior-first unsupported as a deployable scaffold.

The prior-blind and likelihood-only controls serve the complementary purpose. They remove information channels and expose whether a model can answer from incomplete evidence. If those controls performed as well as the full-information prompts, the benchmark would be suspect because answers could be obtained without the manipulated prior or likelihood. Their lower scores support the interpretation that information availability matters, even though they do not by themselves prove Bayesian reasoning.

The scoring fields are deliberately output-level. The scorer does not infer hidden model beliefs, inspect activations, or assign credit for unobserved reasoning. It records what the response displays: option choice, mapped hypothesis, posterior expression, arithmetic consistency, extraction quality, truncation quality, and format compliance. This output-level design keeps the paper aligned with the evidence it has. It can report parser-clean arithmetic and posterior-consistency failures; it cannot report internal causal mechanisms.

The same construction principle explains the treatment of external benchmarks. A public cognitive-bias benchmark can be valuable even when it lacks all schema-v2 fields, but the claim it supports must match the fields it contains. If it supplies only final answer labels, it can support an answer-choice

comparison. If it also supplies posterior targets or probability expressions, it may support posterior-consistency analysis. If it lacks paired prior variants, it cannot support the same prior-flip sensitivity claim without additional construction.

This row-construction policy keeps future work from weakening the current result. A harder split should be built by changing prior and likelihood values while preserving paired structure, not by selecting failures after seeing model outputs. A model sweep should keep prompt definitions and parser fields stable before aggregating across backends. A new scaffold should run on the same rows as the baselines before being evaluated on a new task family. These requirements make future extensions slower, but they keep the interpretation tied to the manipulated variable.

The policy also explains why the paper does not average all rows into one headline score. The hosted rows and Qwen rows answer different regime questions. The hosted rows show ceiling behavior for the active split. The Qwen rows expose non-ceiling, parser-clean failures. Averaging them would obscure both conclusions and could make prior-first look better or worse for the wrong reason. The paper therefore reports the matrix and paired comparisons rather than a single global score.

Finally, the construction notes clarify the denominator. The paper reports full-run rows, not cherry-picked successful parses or smoke-test examples. The active matrix contains 840 scored rows across two backends and seven conditions. Every reported condition cell uses a fixed denominator, and the diagnostic rows record quality gates separately. This is the reason the paper can treat a measured negative as informative: the failure is tied to completed rows, paired comparators, and parser-clean diagnostic fields.

Boundary Cases for Future Rows

Several boundary cases are useful for interpreting future evidence. A row with a correct final option and an incoherent posterior should count as correct for base-rate choice accuracy but not for posterior consistency. This distinction

matters because answer choice and probabilistic expression are different observable behaviors. A benchmark that collapses them would make Qwen-like failures look cleaner than they are.

A row with an incorrect final option but a coherent posterior statement should be inspected under the displayed-option and mapping fields rather than automatically treated as probabilistic success. The final option is the user-facing answer. The posterior text can still be useful as a diagnostic field, but it cannot replace the answer-choice metric unless the task is explicitly redesigned as a probability-elicitation task.

A row with a parser-quality failure should be separated from reasoning failure. If extraction or truncation prevents a usable answer, the paper should report the quality gate and avoid interpreting the row as base-rate neglect. The current full rows do not depend on this escape hatch because the reported Qwen diagnostic cells have zero extraction and truncation failures, but the rule is important for future backends that may be less format-stable.

A row from an external benchmark can be useful even when it lacks posterior fields. In that case, the admissible claim is limited to final-choice behavior under that benchmark’s answer format. A posterior-consistency or arithmetic-failure claim would require additional fields or a separate elicitation design. This boundary keeps external validation from diluting the schema-v2 interpretation.

A row from a new scaffold should be compared first on the existing split before being mixed with a new benchmark family. Otherwise, a win could come from task selection rather than the scaffold. Conversely, a harder split should be run with the existing prompt set before being used to advertise a new method. This ordering preserves the causal interpretation of what changed: task difficulty, prompt instruction, backend, or scorer.

These boundary cases are included because they are common places where a measured negative can be softened into an unsupported positive claim. The present paper avoids that

move. It keeps posterior-only as a control, prior-first as a tested but unsupported candidate, external cells as missing evidence, and output-field diagnostics as observations rather than mechanism claims. A future revision can change those statuses only by adding the corresponding scored rows and refreshed comparisons.

Scope, Literature, and Claim Boundaries

The appendix keeps scope control in one place. The evidence covers one benchmark family, two evaluated backends, seven prompt conditions, and one candidate prior-first scaffold. That is enough to reject the active prior-first superiority claim under the tested protocol, but not enough to characterize all base-rate tasks, all hosted or open-weight systems, or all train-free scaffolds.

The benchmark inherits its central manipulation from classic base-rate work: the semantic item and likelihood evidence stay fixed while the prior changes (Tversky and Kahneman 1974; Bar-Hillel 1980). The frequency-table baseline is included because natural-frequency representations are a substantive train-free Bayesian-reasoning comparator rather than a cosmetic prompt variant (Gigerenzer and Hoffrage 1995). Recent work on LLM belief, posterior expression, and Bayesian elicitation motivates recording final answer, posterior consistency, and arithmetic fields separately (Li and Rui 2024; Freedman and Toni 2025; Hobor et al. 2026).

Cognitive-bias and probabilistic-reasoning benchmarks motivate external validation, but they do not supply result rows for this paper (Echterhoff et al. 2024; Shaikh et al. 2024; Malberg et al. 2024; Zhu, Yan, and Griffiths 2025). Prompting and Bayesian-teaching studies motivate the original train-free intervention question (Araya 2025; Qiu et al. 2025; Sun et al. 2025). The completed matrix answers the narrower question it ran: prior-first prompting does not beat the strong train-free baselines in the active schema-v2 split.

Table 4 also records the claim-status boundary. Supported statements are local regime and diagnostic statements: API ceiling on the active split, Qwen non-ceiling arith-

metic/posterior failures, and posterior-only as an information-availability control. The rejected statement is prior-first superiority under the active paired protocol. Missing statements remain external benchmark generalization, broad model-family scope, and internal mechanism claims.

External-Validation Requirements

A future external cell should preserve five fields when the source benchmark permits them: displayed option, Bayesian hypothesis, posterior expression, arithmetic or calculation consistency, and parser-quality status. If an external cognitive-bias benchmark has only final multiple-choice labels, it can contribute a final-choice result, but it cannot support posterior-consistency or arithmetic-failure claims without schema adaptation.

The evidence package named in Table 5 is deliberately short in the cells and explicit in prose. An external benchmark needs access and use checks, a task filter, raw outputs, scored labels, and compatible aggregates before it becomes a result cell. A harder internal split needs prespecified items, paired prior variants, balanced displayed-option sides, balanced Bayesian-hypothesis sides, and exact paired tests. Difficulty should not come from parser-hostile wording or from selecting failures after the fact.

For a new scaffold, the paper-facing report should specify the intended failure target before reporting wins. A scaffold aimed at prior attention should improve paired prior sensitivity and final base-rate choice accuracy. A scaffold aimed at computation should improve arithmetic failure and posterior numeric consistency. In either case, the scaffold should be compared against direct, generic reasoning, and frequency-table prompting under exact paired tests.

Supplementary Diagnostic Tables

The final supplementary table exposes the row fields behind the main claims without adding new conclusions. Table 6 expands the Qwen diagnostics because the non-ceiling backend is where failure interpretation matters.

Table 6 is placed with this subsection because it explains the same non-ceiling regime discussed in the surrounding prose. The table is not a new benchmark cell. It is a compact expansion of the Qwen rows already summarized in the body: parser-quality gates remain zero, while arithmetic failure and posterior-consistency fields explain why the final-choice score stays low.

Schema Field Roles and Minimal-Pair Validity

A prior-flip item must preserve the semantic situation and likelihood evidence across variants while changing the prior enough that the Bayesian hypothesis can change. If likelihood wording changes with the prior, the row no longer isolates base-rate use. If displayed answer format changes with the prior, the row can measure option-side sensitivity instead of prior sensitivity.

The longer rationale sits outside the table. A model can choose option A, but option A means different hypotheses in different rows; the mapping field prevents side preference from being counted as a Bayesian judgment. Posterior numeric consistency is a further check rather than a replacement for final-choice accuracy: some responses choose the correct option while giving a posterior number that is incompatible with the prior and likelihood evidence. Parser-quality fields remain separate so extraction or truncation artifacts do not become reasoning claims.

The hosted backend and Qwen backend are not averaged into one headline because they occupy different regimes. The hosted backend shows that the split can be solved under several ordinary prompts, but it cannot show a prior-first gain when direct and generic reasoning already reach the same ceiling. Qwen is non-ceiling while still producing parseable, non-truncated answers, so its arithmetic and posterior-consistency fields are informative for failure analysis. Keeping the regimes separate prevents the hosted ceiling from hiding the negative Qwen comparison.

Table 5: External-cell readiness summary: benchmark names remain future validation targets until the listed evidence package exists.

Target cell	Mapping need	Evidence package	Status
CANNOT-style task	label; family; options	source; outputs; labels	motivation only
BiasBuster-style task	category; label; prompt	filter; manifest; aggregates	motivation only
Bayesian Teaching cell	instruction; posterior	matched rows; comparators	design cue
Harder schema-v2 split	active row fields	prespecified pairs; tests	future split
Model sweep	prompts; parser fields	budgets; extraction; aggregates	future validation

Table 6: Qwen diagnostic matrix: arithmetic and posterior-consistency fields explain the non-ceiling regime; extraction and truncation failures are zero for every row.

Condition	n	Acc.	Post. ok	Arith. fail
Direct	60	28.3%	23.3%	76.7%
Generic CoT	60	28.3%	35.0%	65.0%
Frequency table	60	33.3%	21.7%	78.3%
Prior first	60	28.3%	20.0%	80.0%
Posterior only	60	55.0%	0.0%	100.0%
Prior blind	60	31.7%	1.7%	98.3%
Likelihood only	60	25.0%	1.7%	98.3%

Supplementary Slice Checks

The grouped diagnostic slices add one more check on the negative result without changing the claim scope. They ask whether the row matrix is dominated by parser loss, one prompt name, one displayed side, or one prior direction. The parser-quality answer is already clear: extraction, truncation, and format-noncompliance rates are zero in the reported full rows. The remaining variation therefore comes from answer choice, posterior expression, and arithmetic fields rather than from unusable responses.

The option-side slice is useful because the dataset balances displayed answer sides, but a model can still show side sensitivity. On Qwen, rows where the correct displayed option is A have 26 correct answers out of 210, or 12.4% accuracy. Rows where the correct displayed option is B have 112 cor-

rect answers out of 210, or 53.3% accuracy. That asymmetry is not reported as a new benchmark result; it is a warning for interpretation. It explains why the paper keeps the mapped Bayesian-hypothesis field next to the displayed-option field and why paired prior variants are needed before judging a prompt condition.

The prior-direction slice gives a related warning. On Qwen, rows whose pair direction maps to high-A have 27 correct answers out of 210, or 12.9% accuracy. Rows whose pair direction maps to high-B have 111 correct answers out of 210, or 52.9% accuracy. The hosted backend is much less diagnostic on this slice because many rows are already saturated. The Qwen pattern instead supports the paper’s narrow language: the backend exposes a non-ceiling regime with directional answer behavior, but it does not justify a broad statement about all open-weight systems or all base-rate tasks.

Table 7: Schema-v2 field summary: separate recorded fields keep answer choice, posterior wording, and parser quality from collapsing into one score.

Field	Recorded signal	Reason kept separate	Diagnostic role
Displayed option	selected answer	side versus hypothesis	choice accuracy
Bayesian hypothesis	mapped category	row-specific option map	prior sensitivity
Posterior expression	stated probability	answer versus probability	consistency check
Arithmetic flag	incompatible calculation	computation failure	Qwen analysis
Extraction/truncation	missing or cut off answer	parser quality	zero-failure gate

Dose-level slices check whether the negative prior-first conclusion is merely an artifact of one difficulty setting. In the Qwen full rows, the grouped base-rate choice accuracy is 35.7% for the extreme slice, 33.9% for the medium slice, 33.7% for the strong slice, and 33.0% for the weak slice. These values are all far from the hosted ceiling and close enough to each other that the appendix does not promote a dose-level mechanism claim. They do, however, support the decision to treat Qwen as the failure-analysis backend rather than as a single anomalous condition row.

The same grouped table also separates posterior consistency from final choice. Across Qwen rows with posterior-numeric consistency marked true, base-rate choice accuracy is 31/62, or 50.0%. Across rows marked false, accuracy is 107/358, or 29.9%. This association is useful but not causal evidence: the run is an output audit, not an intervention on hidden computation. The paper therefore uses it to motivate the failure taxonomy and future scaffold requirements, while keeping the headline result anchored to the condition matrix and exact paired tests.

Reproducibility and Update Semantics

The manuscript reports reproducibility at the level needed to evaluate the scientific claims: task family, row count, prompt conditions, evaluated backends, deterministic budgets, parser fields, aggregate metrics, and paired tests. It intentionally avoids local execution details in reader-facing prose because those details are not part of the benchmark definition or result

interpretation.

Future evidence updates follow a fixed synchronization rule. A harder split would require refreshed aggregate tables, paired tests, failure taxonomies, and captions, and it would unlock a stronger claim only if paired prior variants and balanced sides are preserved. A new prompt scaffold would need the same split, direct/generic/frequency comparators, and paired tests before it could support a mitigation claim. A new backend would need clean extraction, comparable budgets, and per-backend aggregates before it could broaden model scope. An external benchmark would need a source record, schema map, raw outputs, and scored rows before it could become a result cell. A parser revision would require row rescoring and refreshed quality gates before any affected figure or table could be reused.

The scientific unit of reproduction is a scored row matrix. A reproducer should be able to rebuild prior-flip rows, run the evaluated backends or comparable backends under declared budgets, parse responses into schema-v2 fields, and regenerate the condition matrix and paired tests. If a local environment changes but the row matrix and scoring policy are equivalent, the scientific claim is unchanged. If the row matrix or scoring policy changes, the result claims must be refreshed.

The appendix closes with three evidence states. Prior-first superiority is rejected under the active protocol because it ties saturated hosted baselines and trails frequency-table

prompting on Qwen. Posterior-only is supported only as an information-availability control. External benchmark families, harder splits, and model sweeps are missing validation cells until completed scored rows exist.

This distinction keeps the negative result readable without inflating it. A reader can inspect Table 1 for condition-level denominators, Table 2 for matched comparisons, Figure 4 for failure fields, and Table 6 for the Qwen diagnostic expansion. The practical reading order is therefore short: confirm the task scale and evaluated backends in the setup, inspect the two result tables for the prior-first comparison, inspect the Qwen failure figure and diagnostic table for parser-clean failure fields, and treat the missing-cell ledgers as future-evidence requirements rather than hidden results.

Reader Verification Notes

The supplementary reading path is intentionally evidence-first. The setup table records the size and backend mix of the completed matrix. The condition ledger explains why prior-first is compared against frequency-table prompting rather than against direct prompting alone. The matched-comparison ledger explains why the paper uses paired tests before admitting a method claim. The diagnostic-field ledger explains why Qwen failures are discussed as arithmetic and posterior-consistency failures rather than parser failures.

A reader who wants to verify the negative prior-first conclusion should begin with the Qwen non-ceiling rows. The relevant comparison is not whether prior-first sometimes performs above chance, but whether it beats the strongest fair train-free comparator under matched rows. Frequency-table prompting is the comparator that matters because it changes the representation of Bayesian information without training and is motivated by natural-frequency results in human Bayesian reasoning (Gigerenzer and Hoffrage 1995). The active Qwen rows do not support a prior-first win over that comparator.

A reader who wants to verify the ceiling claim should instead inspect the hosted rows. Direct prompting, generic

reasoning, prior-first prompting, and posterior-only all reach the same ceiling. That pattern supports a regime claim: the active split is solvable for the hosted backend under several ordinary prompts. It does not support a method claim, because a method claim requires separation from its comparator rows.

A reader who wants to verify the schema-v2 contribution should inspect how the scorer separates output fields. The displayed option, mapped hypothesis, posterior expression, arithmetic consistency, extraction quality, and truncation quality fields answer different questions. This separation follows recent warnings that LLM explanations and probabilities can appear coherent while failing probabilistic consistency checks (Li and Rui 2024; Freedman and Toni 2025; Hobor et al. 2026). The paper therefore avoids treating a fluent rationale as sufficient evidence for Bayesian reasoning.

A reader who wants to verify the scope boundary should inspect the missing-cell ledger and external-readiness table. CANNOT-style, BiasBuster-style, Bayesian Teaching, harder-split, and model-sweep cells are scientifically relevant because cognitive-bias behavior can vary with task framing and benchmark source (Echterhoff et al. 2024; Shaikh et al. 2024; Malberg et al. 2024). They are not reported as results here because the active artifact set contains no completed scored rows from those cells.

The practical consequence is that the paper has three allowed endings and one disallowed ending. It may say that prior-first superiority is rejected under the active paired protocol. It may say that posterior-only is a positive information-availability control. It may say that Qwen failures are parser-clean and concentrated in arithmetic/posterior fields. It may not say that all LLMs neglect base rates, that all scaffolds fail, or that external cognitive-bias benchmarks would follow the same pattern.

Design Rationale Notes

The prior-flip construction is the part of the benchmark that most directly connects the experiment to the base-rate litera-

ture. Classic work frames base-rate neglect as a case where salient individuating evidence can dominate prior information (Tversky and Kahneman 1974; Bar-Hillel 1980). The paper translates that setup into a minimal-pair protocol: the semantic item and likelihood evidence remain fixed, and the prior changes. This is why the paper treats a row pair, not a single multiple-choice item, as the meaningful unit for prior sensitivity. A single row can show whether the final answer is correct; a paired row can show whether the answer moves when only the prior changes.

The displayed-option field exists because prior-flip pairs can otherwise be contaminated by answer-side effects. If the Bayesian answer were always option A, a model could appear base-rate sensitive while merely preferring a side. If the same named hypothesis were always the base-rate-sensitive answer, a category association could substitute for probabilistic judgment. The schema therefore records the displayed option and the mapped Bayesian hypothesis separately. This does not make the benchmark broader, but it makes the narrow comparison cleaner: the scorer can distinguish choosing a letter from choosing the hypothesis implied by the prior and likelihood values.

The frequency-table baseline is included for a methodological reason, not as a decorative prompt. Natural-frequency representations have a long role in Bayesian-reasoning research because they can change how people represent conditional information without requiring formal probability notation (Gigerenzer and Hoffrage 1995). For this paper, that literature makes frequency-table prompting a serious train-free comparator. A prior-first instruction is not credited merely because it sounds more Bayesian than a direct prompt. It must separate from a frequency representation that also changes the inference format without training.

The posterior and arithmetic fields are included because fluent explanations can be misleading evidence. Recent LLM work on belief, posterior expression, and Bayesian elicitation motivates the concern that a model can write a plausible

rationale while failing to maintain a coherent probability judgment (Li and Rui 2024; Freedman and Toni 2025; Hobor et al. 2026). The Qwen rows make that concern concrete: many outputs are parseable and complete, but the posterior expression or arithmetic is incompatible with the item. This is why the paper discusses failure fields rather than reporting only final-choice accuracy.

The parser-quality fields serve the opposite purpose: they prevent a failure-analysis claim from being too easy. If extraction failures or truncation failures explained the wrong answers, the appropriate conclusion would be about response format or generation length, not base-rate reasoning. The active full rows do not show that pattern. Extraction and truncation failures are zero for the reported Qwen diagnostic rows, which allows the paper to interpret the non-ceiling regime as parser-clean output behavior while still avoiding hidden-mechanism claims.

The hosted-backend ceiling is a result about the active split, not a general statement about hosted models. Direct prompting, generic reasoning, prior-first, and posterior-only all reach ceiling on the hosted rows. That pattern is useful because it shows that the items are solvable, but it also limits the intervention claim: a scaffold cannot be credited with improving a comparator that already obtains every item. The paper therefore separates task solvability from method superiority. This separation is the main reason the abstract reports a negative prior-first comparison rather than a positive mitigation result.

The Qwen backend contributes a different kind of evidence. It is non-ceiling, parseable, and run on the same prompt-condition family, so it supplies a regime where prompt comparisons and failure fields matter. In that regime, frequency-table prompting is above prior-first, and posterior-only is strongest because it changes information availability. The paper does not average this with the hosted ceiling rows because averaging would hide both facts: the hosted split is saturated, and the Qwen split does not support prior-first superiority.

The external-benchmark discussion is deliberately separated from the result table because benchmark scope can change the phenomenon being measured. Cognitive-bias evaluations vary in task source, framing, response format, and bias taxonomy (Echterhoff et al. 2024; Shaikh et al. 2024; Malberg et al. 2024). A benchmark name alone does not guarantee compatibility with schema-v2 fields. A public benchmark cell becomes a result only when source access, task filtering, raw outputs, scoring fields, and compatible metrics exist. Until then, those benchmarks are motivation and future validation targets, not hidden baselines.

The same caution applies to probability-focused benchmarks and representation studies. Work on recovering event probabilities from model representations motivates probability-aware evaluation, but it does not replace completed prior-flip rows in this paper (Zhu, Yan, and Griffiths 2025). A hidden-state or representation result would answer a different question from the output-field protocol used here. The present paper can say that Qwen outputs show arithmetic and posterior-consistency failures; it cannot say that the model internally uses or ignores a prior unless a different experiment measures that mechanism.

Prompt-scaffold and Bayesian-teaching studies motivate the intervention question while also showing why comparator discipline matters (Araya 2025; Qiu et al. 2025; Sun et al. 2025). An instructional or scaffolded prompt can change behavior, but a paper-facing claim must still name the comparator and the task slice. The active candidate scaffold is prior-first. The active comparators include direct prompting, generic reasoning, and frequency-table prompting. Under that comparator set, the result is negative for prior-first superiority.

The posterior-only control is useful precisely because it is not a deployable answer to the original task. It supplies posterior information rather than asking the model to infer it, so it changes the information available to the respondent. This makes it a positive diagnostic control: the benchmark

can register an information-channel effect when that channel is supplied. Calling posterior-only a mitigation would be wrong, but omitting it would make the negative prior-first result harder to interpret because readers could suspect that the benchmark cannot register any prompt-dependent signal.

The missing-cell policy is also a claim policy. A missing harder split does not show that prior-first would fail on harder items. A missing public benchmark cell does not show that the current schema would transfer. A missing model sweep does not show that the two evaluated backends are representative. The manuscript therefore uses absence only to delimit scope. It does not convert absent rows into negative evidence.

The update policy follows from that same distinction. If a future harder split shows a prior-first win over direct, generic reasoning, and frequency-table prompting, the abstract, main result table, paired tests, failure taxonomy, captions, and conclusion should all change. If a future computation-aware scaffold reduces arithmetic and posterior-consistency failures, the paper should report that scaffold against the same comparator set. The current paper is written so those future results can be added without rewriting the present negative result into a broader claim.

These design choices make the appendix evidence-bearing rather than ornamental. The tables record which cells exist, which fields are interpreted, which claims are admitted, and which extensions remain missing. The prose connects those tables to the cited literature and to the active row matrix. It does not add a new experiment, and it does not make the prior-first scaffold stronger than the completed rows allow.

Evidence Synchronization Notes

The paper is written around synchronized evidence surfaces. The aggregate table gives the condition-level denominators and accuracies. The paired table gives the matched-comparison surface. The Qwen failure figure gives the diagnostic-field decomposition. The supplementary ledgers then explain which of those surfaces admits each claim. This organization matters because a negative intervention result

can otherwise be misread in two opposite directions: as a broad claim that all scaffolds fail, or as a weak result that says nothing because one prompt failed. The synchronized surfaces support the middle claim only.

The abstract’s result sentence is therefore constrained by the tables. It can report that the hosted backend saturates and that Qwen does not support prior-first superiority. It cannot report a broad cross-benchmark finding, because no external benchmark rows are present. It cannot report mechanism discovery, because the recorded evidence is output-field evidence. It cannot report that posterior-only is a deployable method, because posterior-only changes information availability. These exclusions are not stylistic caution; they are consequences of which rows exist.

The introduction follows the same synchronization rule. It can motivate base-rate neglect from classic cognitive work and motivate LLM evaluation from recent probabilistic-reasoning studies (Tversky and Kahneman 1974; Bar-Hillel 1980; Li and Rui 2024; Freedman and Toni 2025). It can explain why frequency representations are a serious comparator (Gigerenzer and Hoffrage 1995). It can identify cognitive-bias benchmark variation as the reason external validation remains important (Echterhoff et al. 2024; Shaikh et al. 2024; Malberg et al. 2024). It cannot imply that those external benchmarks have already been run in this paper.

The method section is synchronized to the scorer fields. A method description that mentioned only final answer accuracy would hide why schema-v2 is useful. A method description that discussed posterior consistency without naming the final-choice metric would make the paper look like a probability-elicitation study rather than a base-rate answer-choice study. The current method needs both: final answer choice to measure the user-visible response, and diagnostic fields to explain how parser-clean failures arise.

The results section is synchronized to comparator discipline. A raw high score is not an improvement claim unless the comparator is named and the matched rows support the

direction. This is why the hosted prior-first row is not a win even though it is numerically perfect: direct and generic reasoning are also perfect. It is also why the Qwen prior-first row is not a win even though direct prompting is low: frequency-table prompting is a stronger train-free comparator on the same backend.

The limitations section is synchronized to missing cells. The paper’s limits are not generic disclaimers about all possible LLM behavior. They name the absent evidence surfaces: external cognitive-bias benchmarks, harder prior-flip splits, broader model sweeps, and internal mechanism probes. Each missing surface would unlock a different kind of claim. External rows would broaden benchmark scope. Harder splits would test the hosted ceiling. Model sweeps would broaden backend scope. Mechanism probes would be needed before discussing hidden computation.

The appendix is synchronized to update discipline. If future rows are added, the condition matrix, paired tests, failure taxonomy, captions, abstract, conclusion, and supplementary ledgers must change together. This prevents the common failure mode in which a result table is updated but the headline story still reflects an older run. The current appendix therefore functions as a claim ledger: it records what the paper is allowed to say now and what evidence would be needed before it could say more.

Future work should add evidence before changing the story. A harder split would test whether the hosted ceiling can be reduced while preserving prior-flip pairing. External cognitive-bias benchmarks would test whether the schema-v2 fields transfer beyond this controlled source family. A computation-aware scaffold would test whether arithmetic and posterior-consistency failures can be reduced. Each extension should retain the same comparator discipline: direct prompting, generic reasoning, frequency-table prompting, and the candidate scaffold must be compared on matched rows before a mitigation claim is admitted.

Supplementary Update Checklist

A future update should begin by naming which evidence cell changed. The cell might be a harder internal split, an external cognitive-bias benchmark, a new backend, or a new prompt scaffold. Each option changes a different claim boundary. A harder split addresses ceiling saturation; an external benchmark addresses source-family scope; a backend sweep addresses model scope; and a prompt scaffold addresses the intervention claim. Mixing these updates without a cell label would make it unclear which conclusion changed.

The first synchronization step is row-level. New rows should preserve the displayed-option field, mapped Bayesian-hypothesis field, posterior-expression field when available, arithmetic flag, extraction flag, truncation flag, and prompt-condition label. If an external benchmark cannot supply one of those fields, the paper should say which field is absent and narrow the dependent claim. A final-choice-only benchmark can still be useful, but it cannot support the same posterior-consistency or arithmetic-failure statements as schema-v2.

The second synchronization step is comparison-level. Any new prior-first or computation-aware scaffold should be paired against direct prompting, generic reasoning, and frequency-table prompting on matched rows before it becomes a mitigation claim. Aggregate accuracy can describe a regime, but paired discordant counts decide whether the candidate condition wins the relevant comparison. This rule is what prevents a high score in a saturated setting from being reported as a method advantage.

The third synchronization step is manuscript-level. If a new evidence cell changes a table value, the abstract, result paragraph, figure caption, paired-test interpretation, limitation statement, and appendix ledger should be regenerated together. If those surfaces disagree, the table and paired-test artifacts are the authority and the prose should be rewritten. This update rule is deliberately conservative because the current paper's value depends on keeping a measured negative from being softened into an unsupported positive claim.

Finally, future work should preserve the distinction between observed output fields and hidden mechanisms. A stronger scaffold may reduce arithmetic failures or improve posterior consistency, but that would still be output evidence unless a separate mechanism experiment is run. The present appendix records this boundary so that a later, stronger result can be added without retroactively changing what the current runs measured.